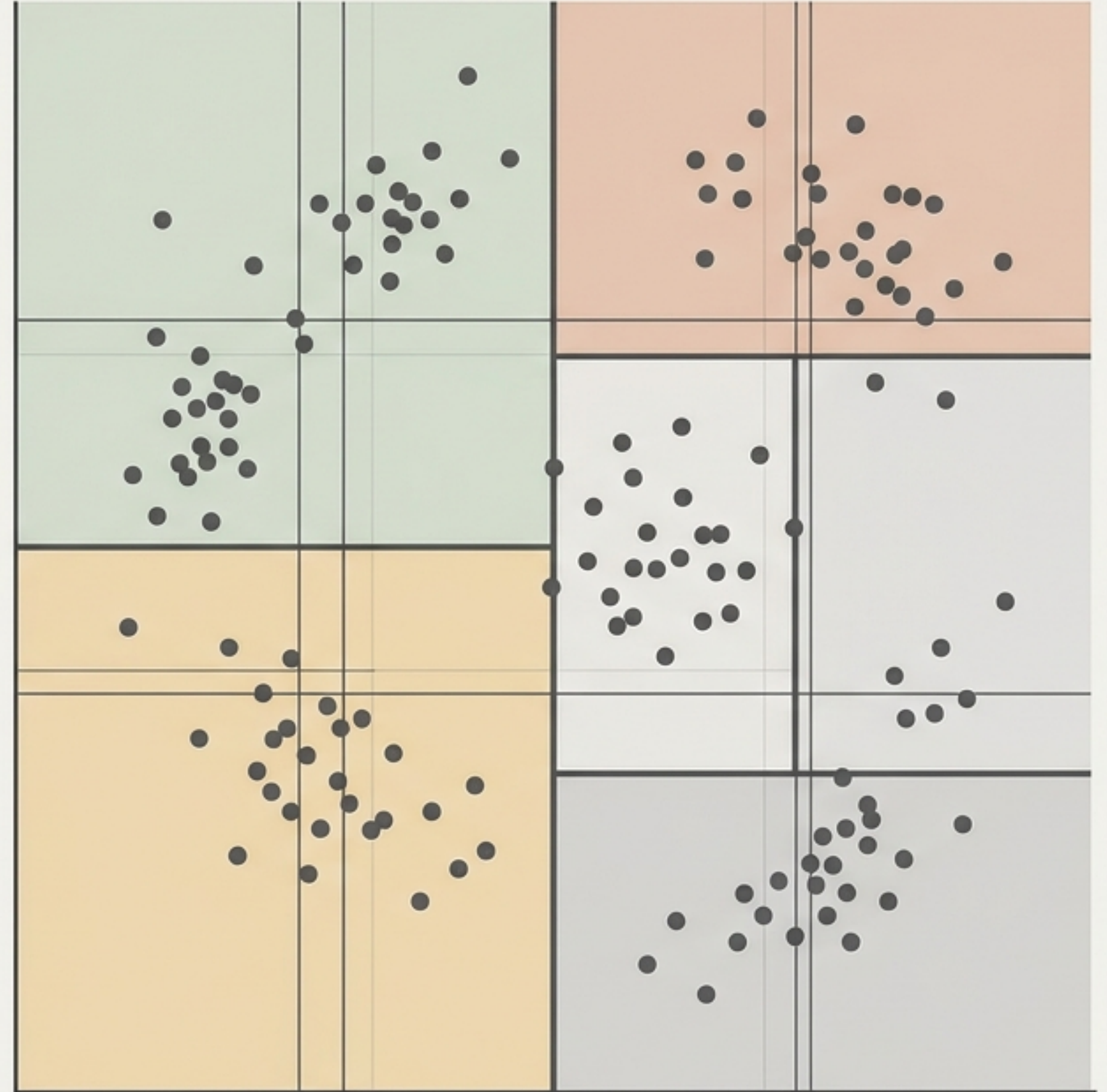


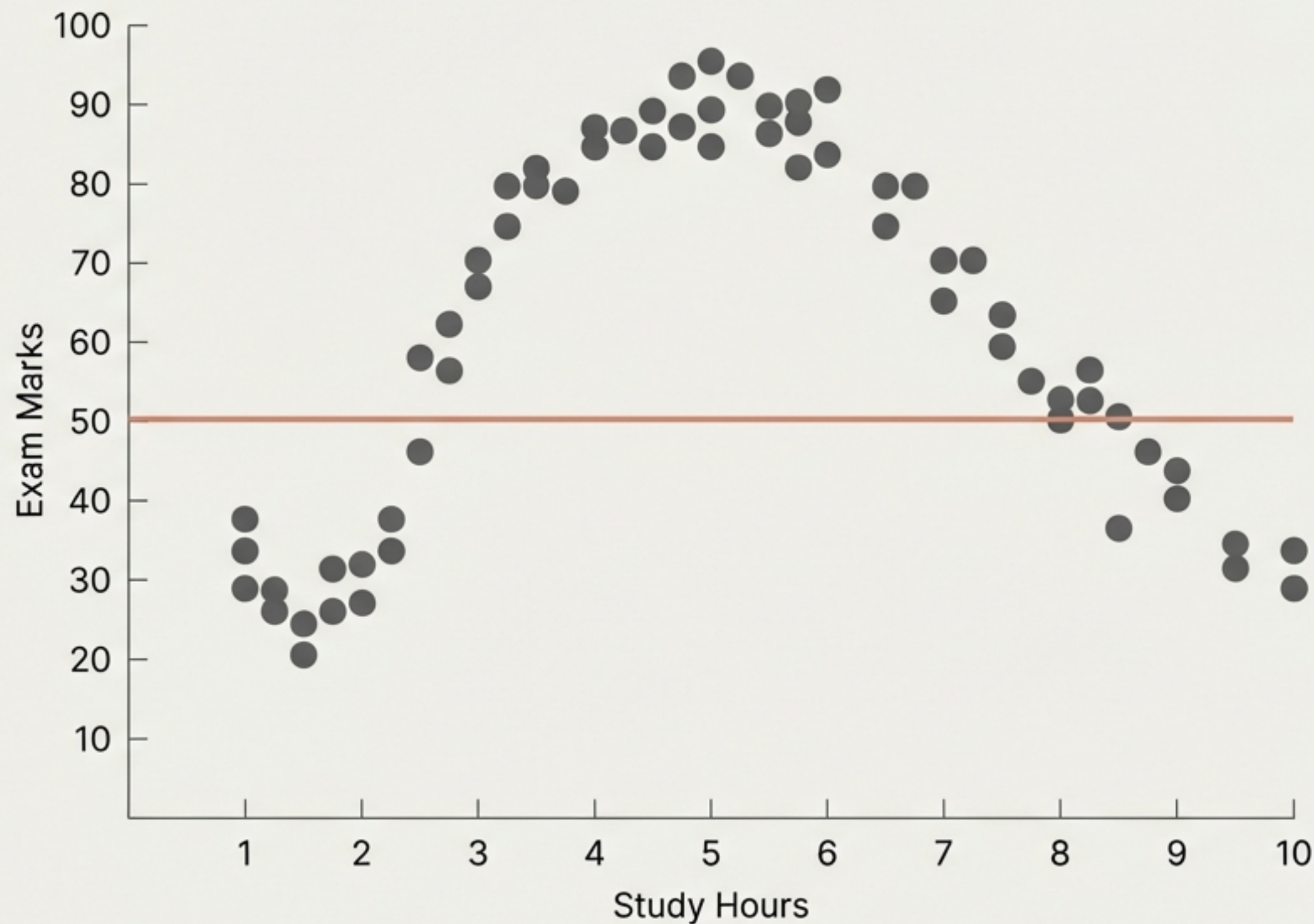
Mastering Regression Trees Trees

From non-linear intuition to
hyperparameter tuning.



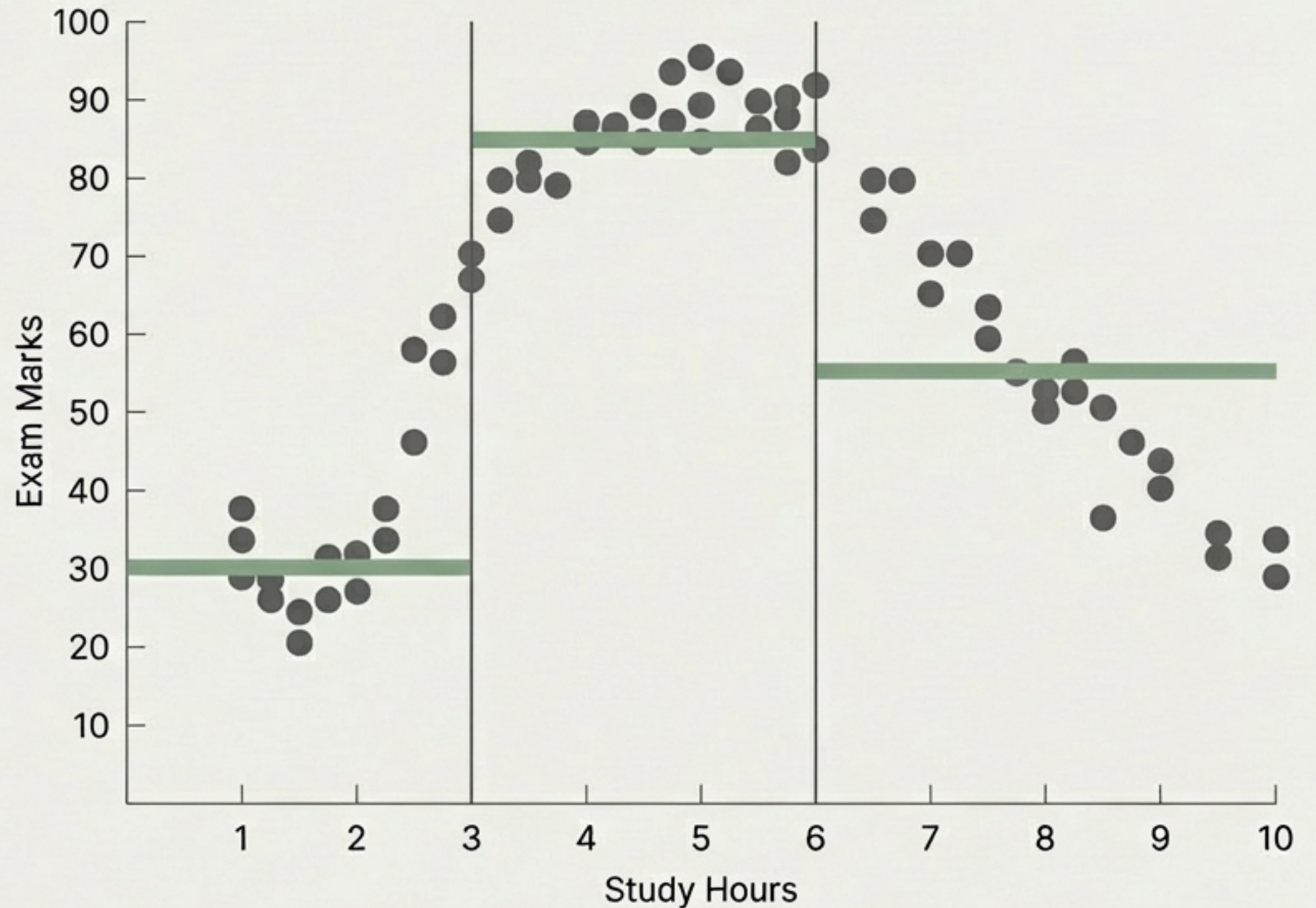
The Limitation of Linear Models

Linear regression attempts a single, continuous equation. On non-linear relationships, it severely underfits.



Slicing the Problem: Enter Regression Trees

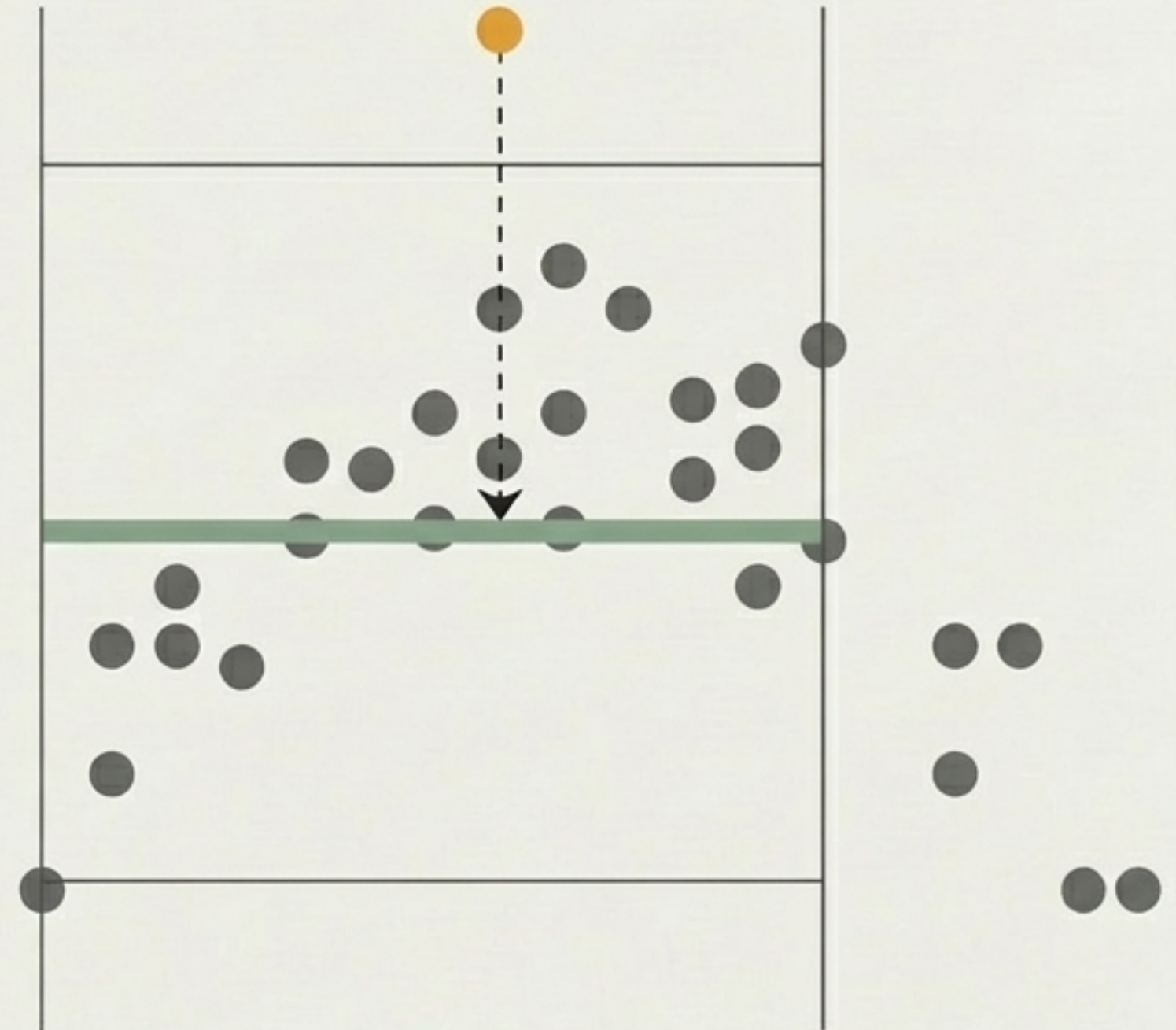
Instead of one continuous line, Regression Trees slice the dataset into distinct logical groups. They don't draw a line; they build a step function.



“The Golden Rule of Prediction” in Playfair Display

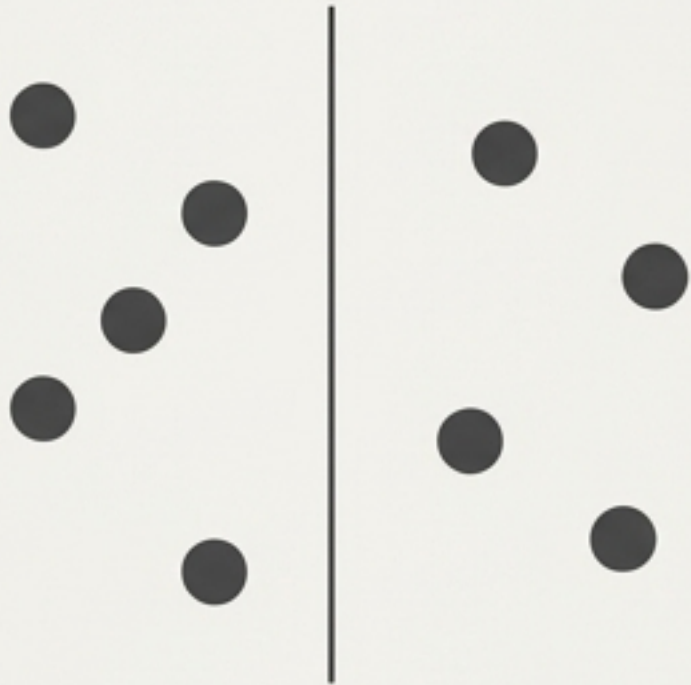
The Rule: The prediction for any new data point is simply the Mean of the target values within its assigned region (leaf node).

In Practice: If a 5-hour student falls into the “3–6 hours” region, their predicted mark is the exact average of all historical students in that same region.

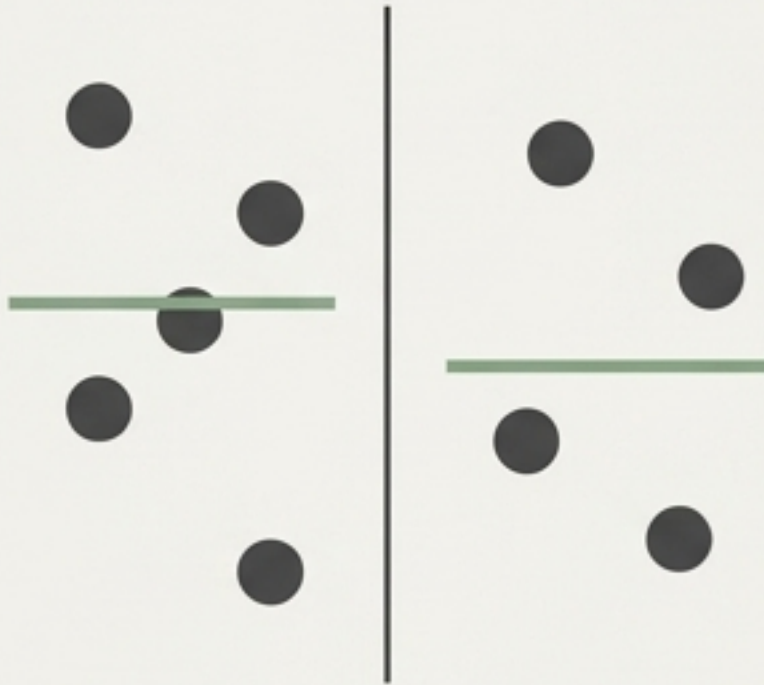


Finding the Perfect Split

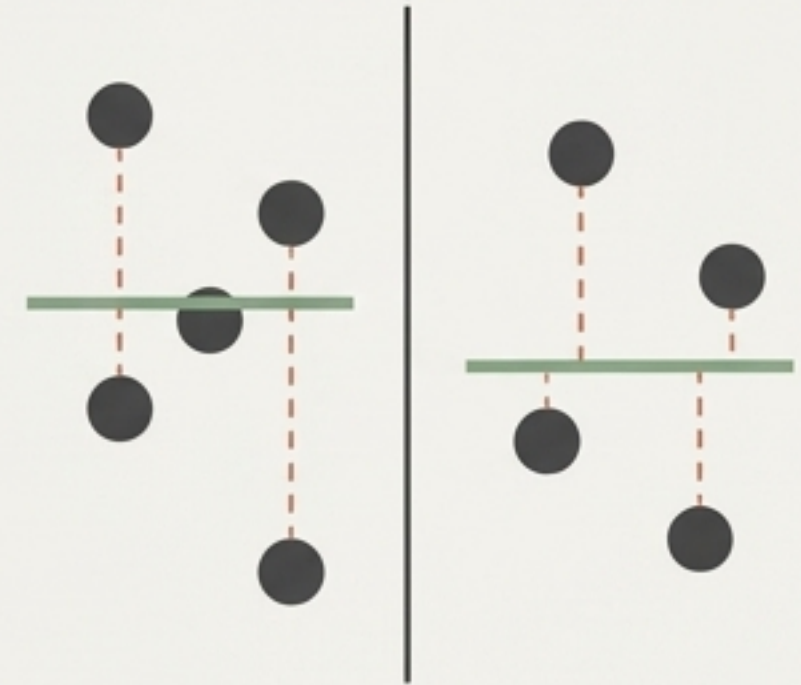
1. The Split



2. The Mean



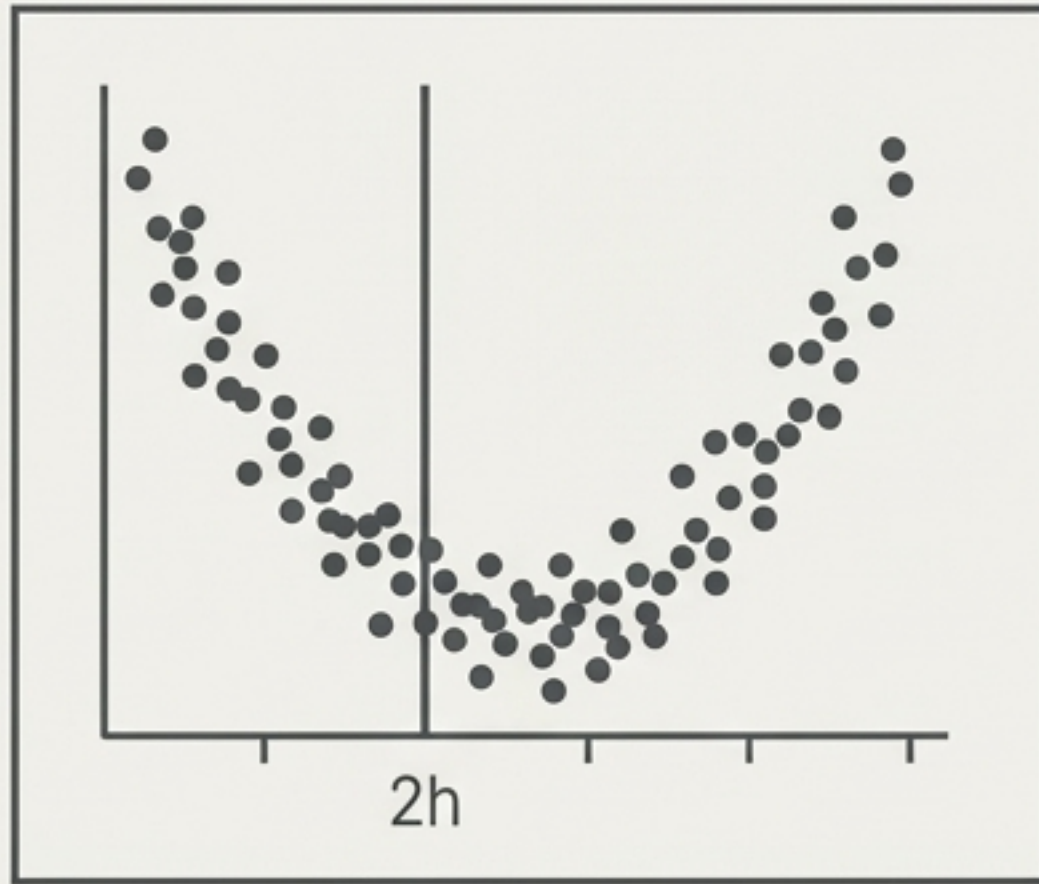
3. The Error



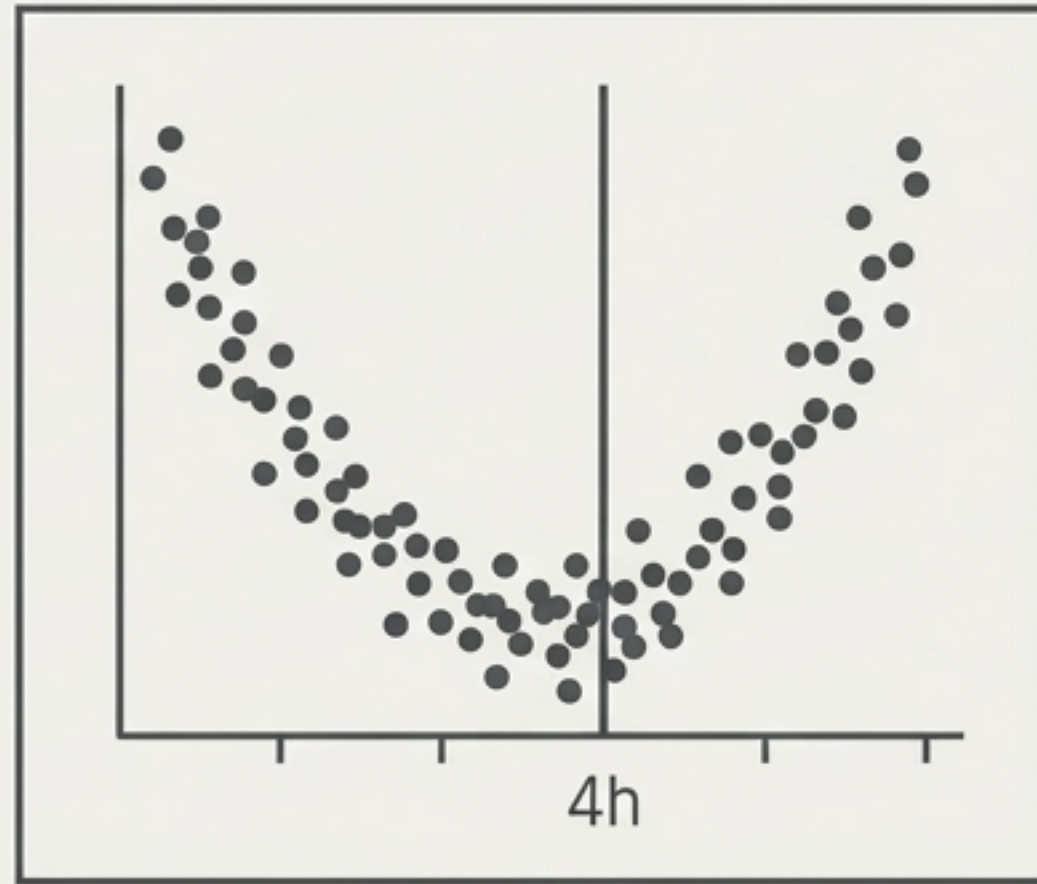
Mean Squared Error (MSE)

The algorithm calculates the distance (error) of each point to the mean, squares it, and sums them up. The lower the total MSE, the better the split.

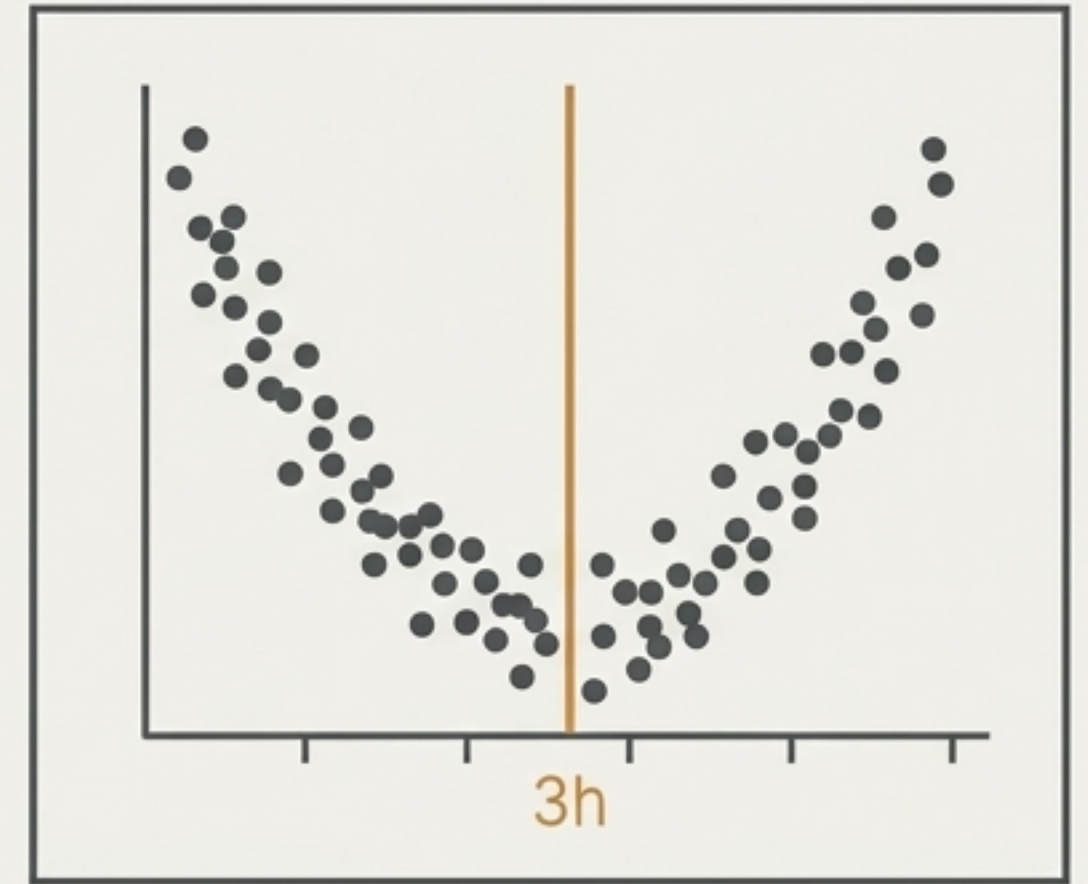
The Iterative Splitting Process



Split at 2 hours
(High MSE)



Split at 4 hours
(Medium MSE)



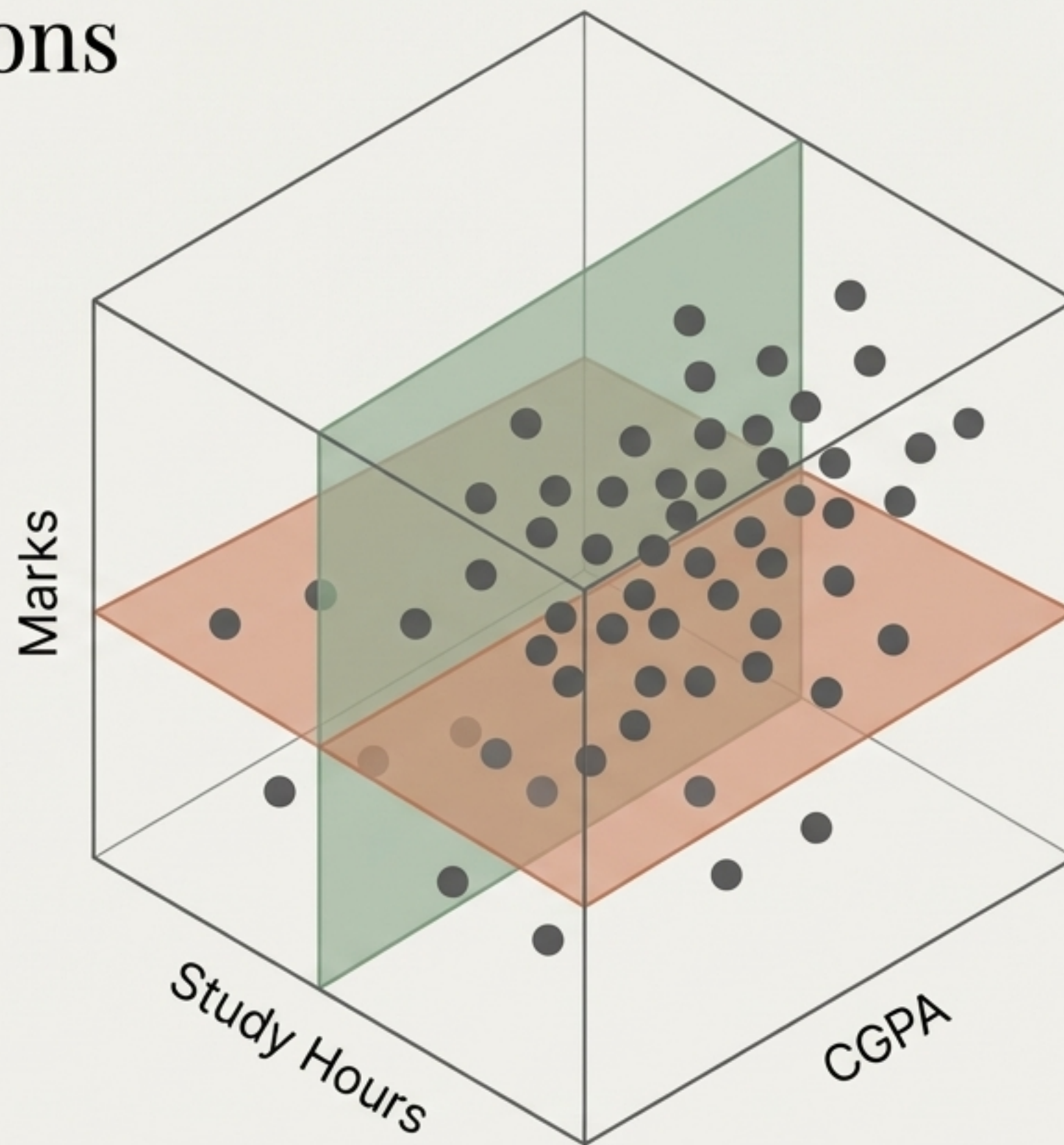
Split at 3 hours
(Minimum MSE)

The algorithm tests multiple cut points across the axis. It evaluates the MSE for every single potential split and locks in the precise location where the total error is at its absolute lowest.

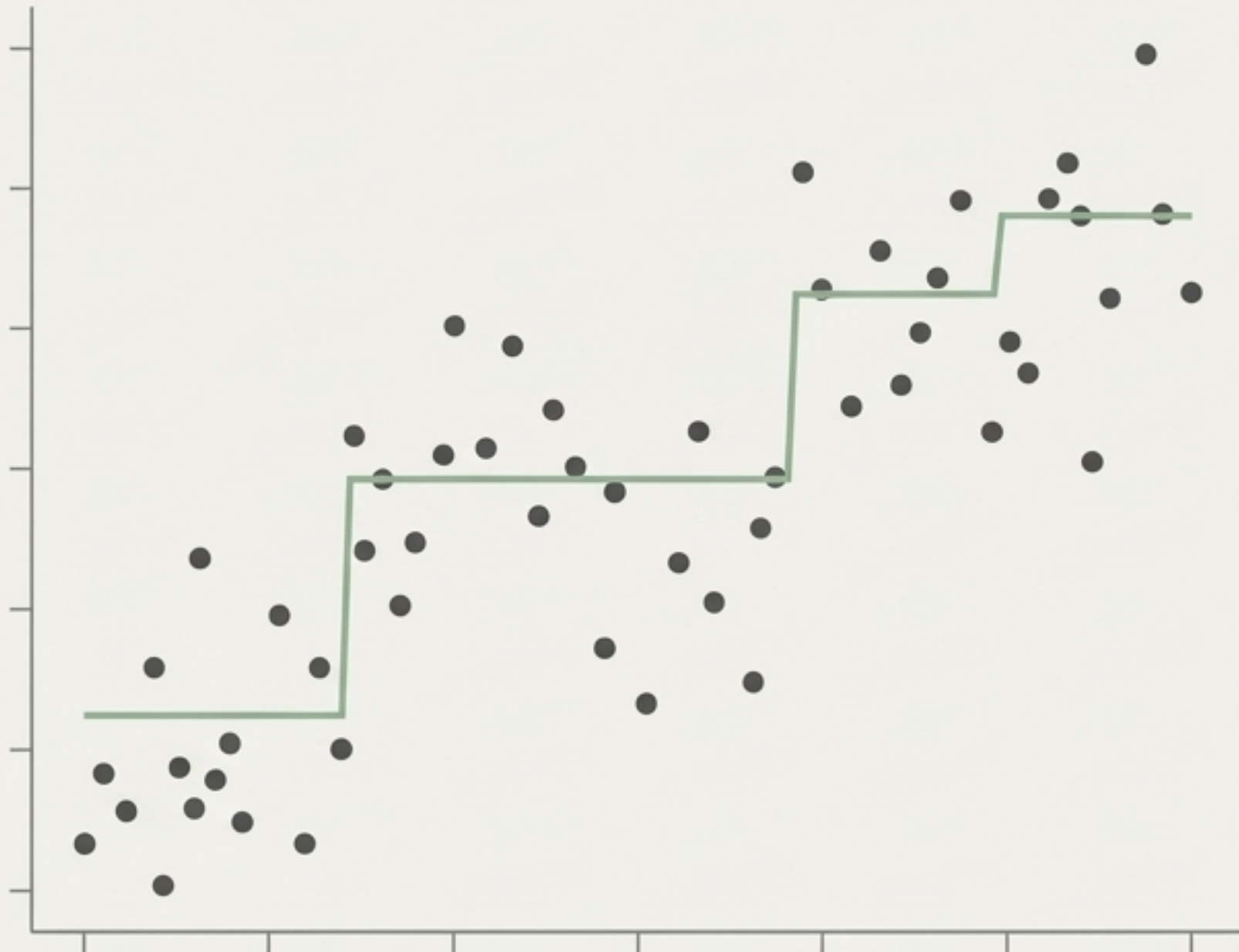
Scaling to Multiple Dimensions

- **Split 1:** Sliced by Study Hours.
- **Split 2:** Sub-sliced by CGPA.

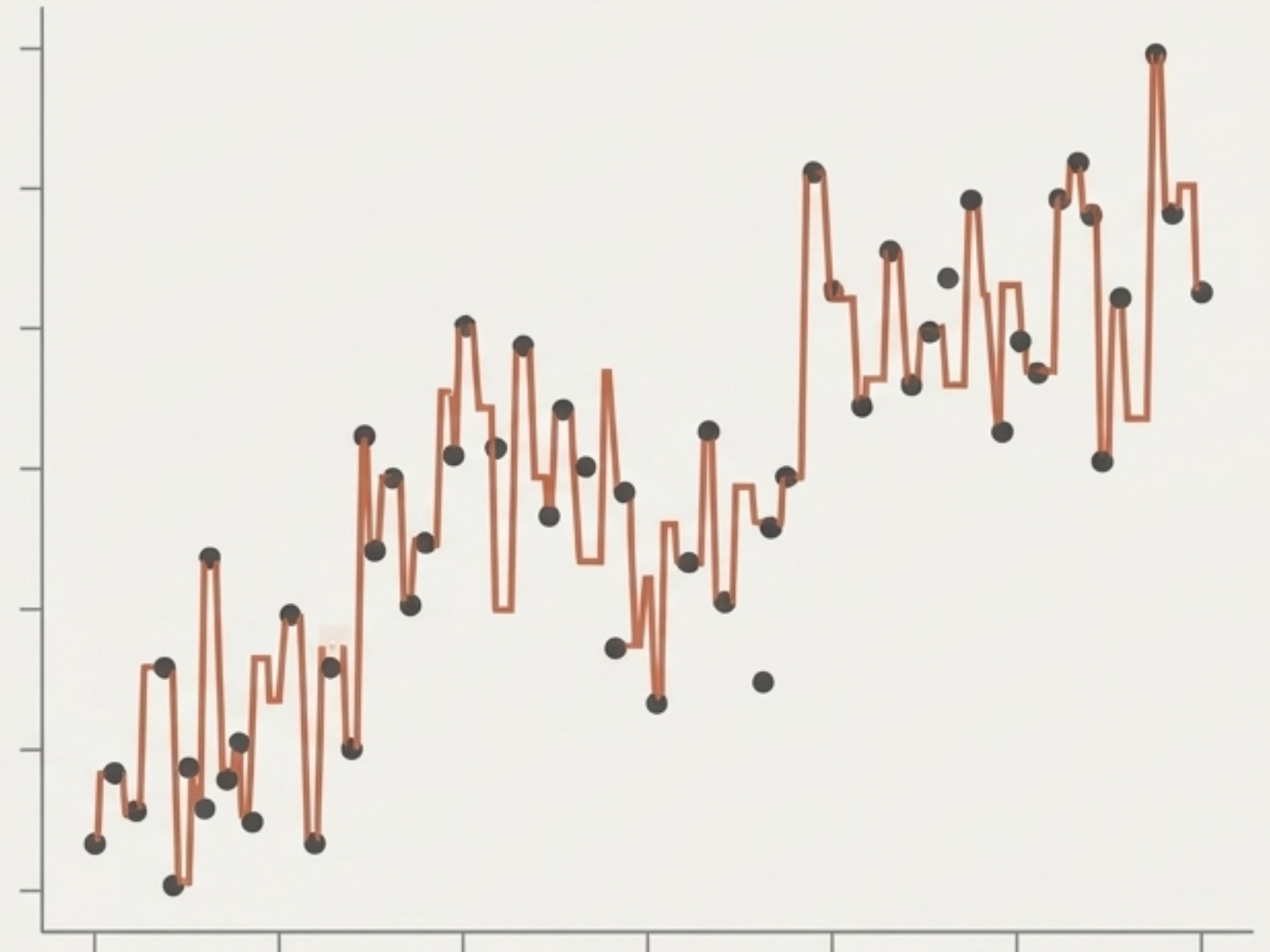
When adding new features (like historical CGPA), splits occur across multiple axes. The algorithm evaluates every variable sequentially, finding the most effective cutting planes to isolate specific clusters.



The Danger of Overfitting



Generalised Model



Overfitted Model (Zero Training Error, Terrible Generalisation)

Left to its own devices, a Regression Tree will keep splitting until every point has its own leaf. We must restrict its growth.

Hyperparameter Toolkit I: Restricting Growth

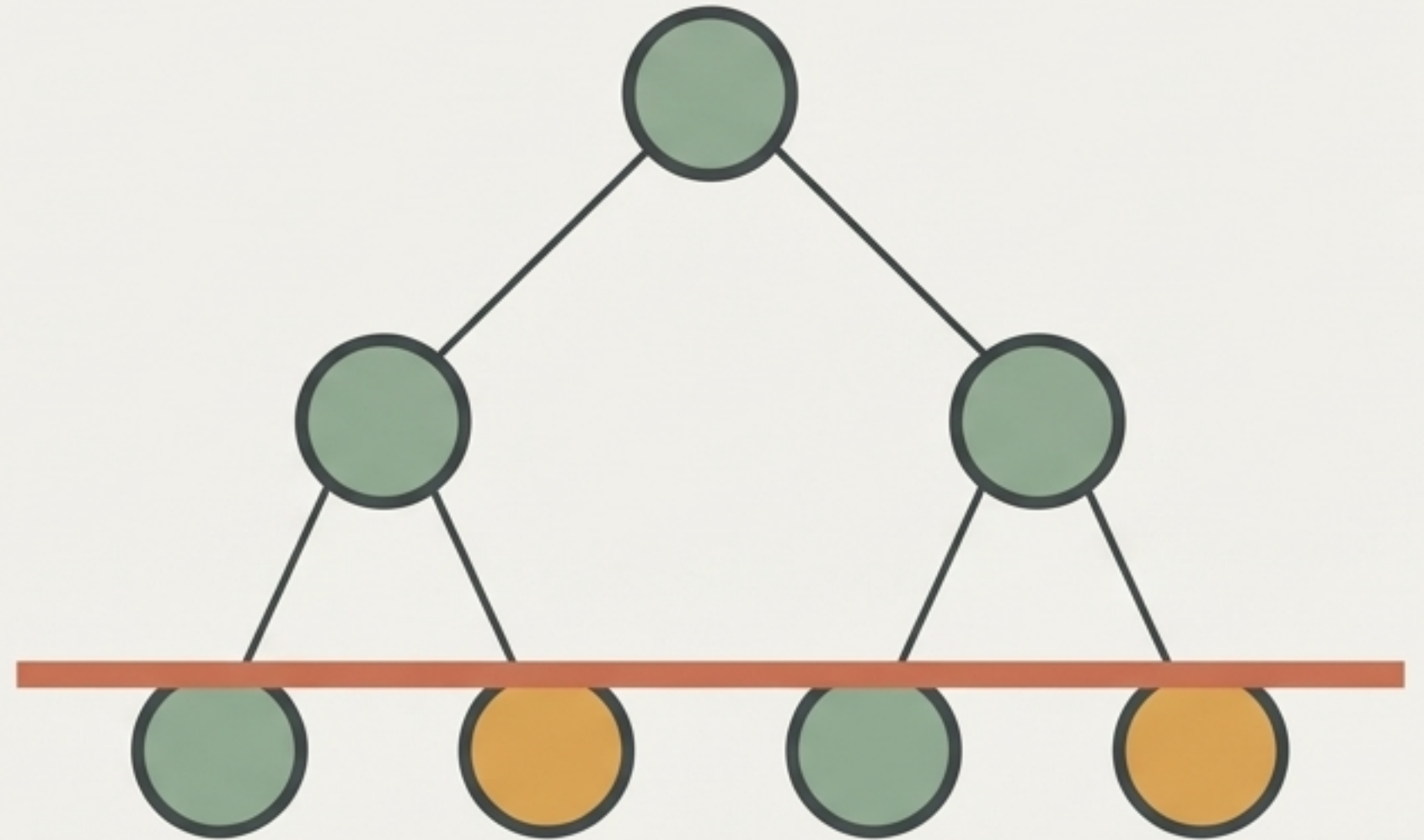
`max_depth`

Caps the vertical levels of the tree.
Depth 50 overfits; depth 5 generalises.

`min_samples_split`



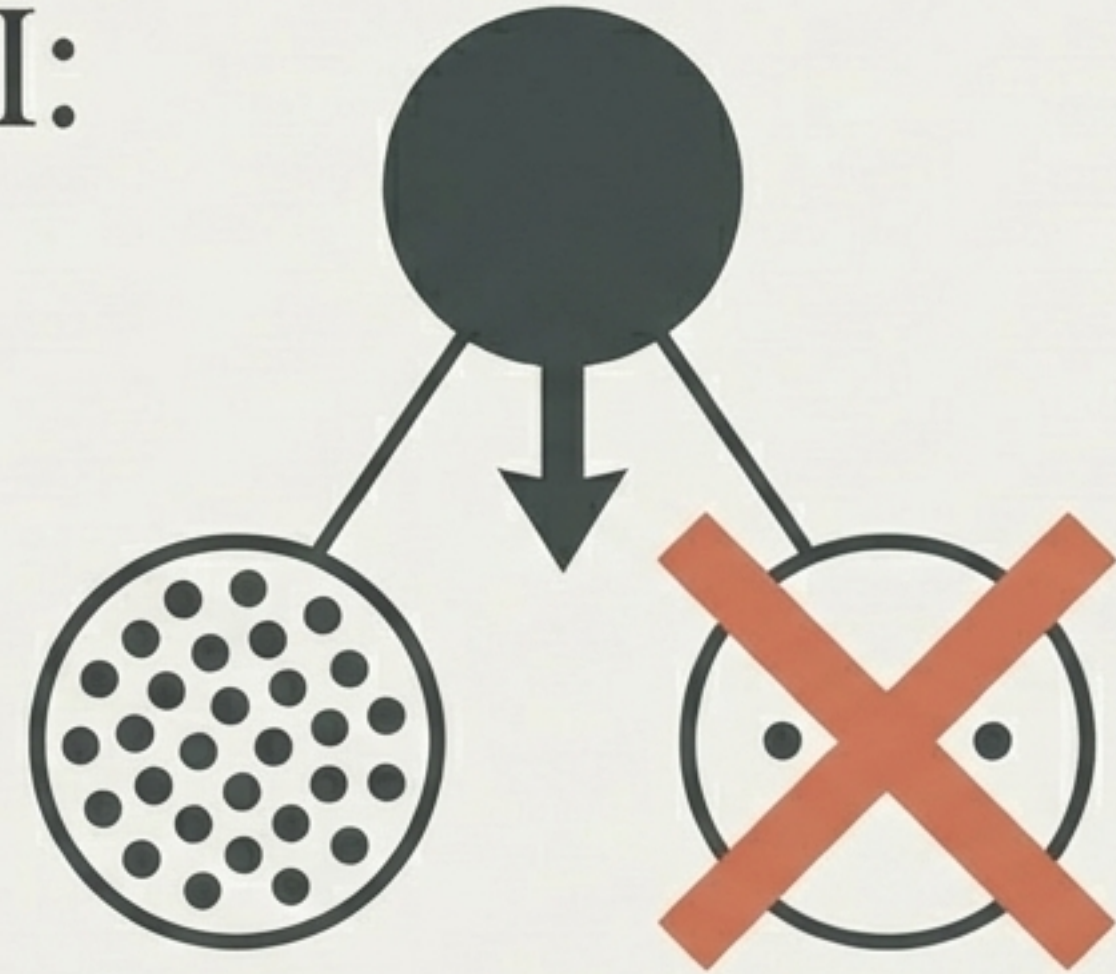
Requires a minimum number of
samples (e.g., 100) before allowing
a node to divide further.



Hyperparameter Toolkit II: Governing Leaves

`min_samples_leaf`

Ensures each final region has enough statistical weight. If a split leaves too few points, it is aborted.



`max_leaf_nodes`

A hard cap on the total number of distinct final regions the model can create.



Hyperparameter Toolkit III: Advanced Levers



The Filter (`min_impurity_decrease`)

Only permits a split if the MSE improvement is substantial. Ignores mathematically negligible splits.



The Randomiser (`splitter = 'random'`)

Instead of evaluating all possible splits to find the absolute best, it picks randomly. Increases speed and helps prevent overfitting.

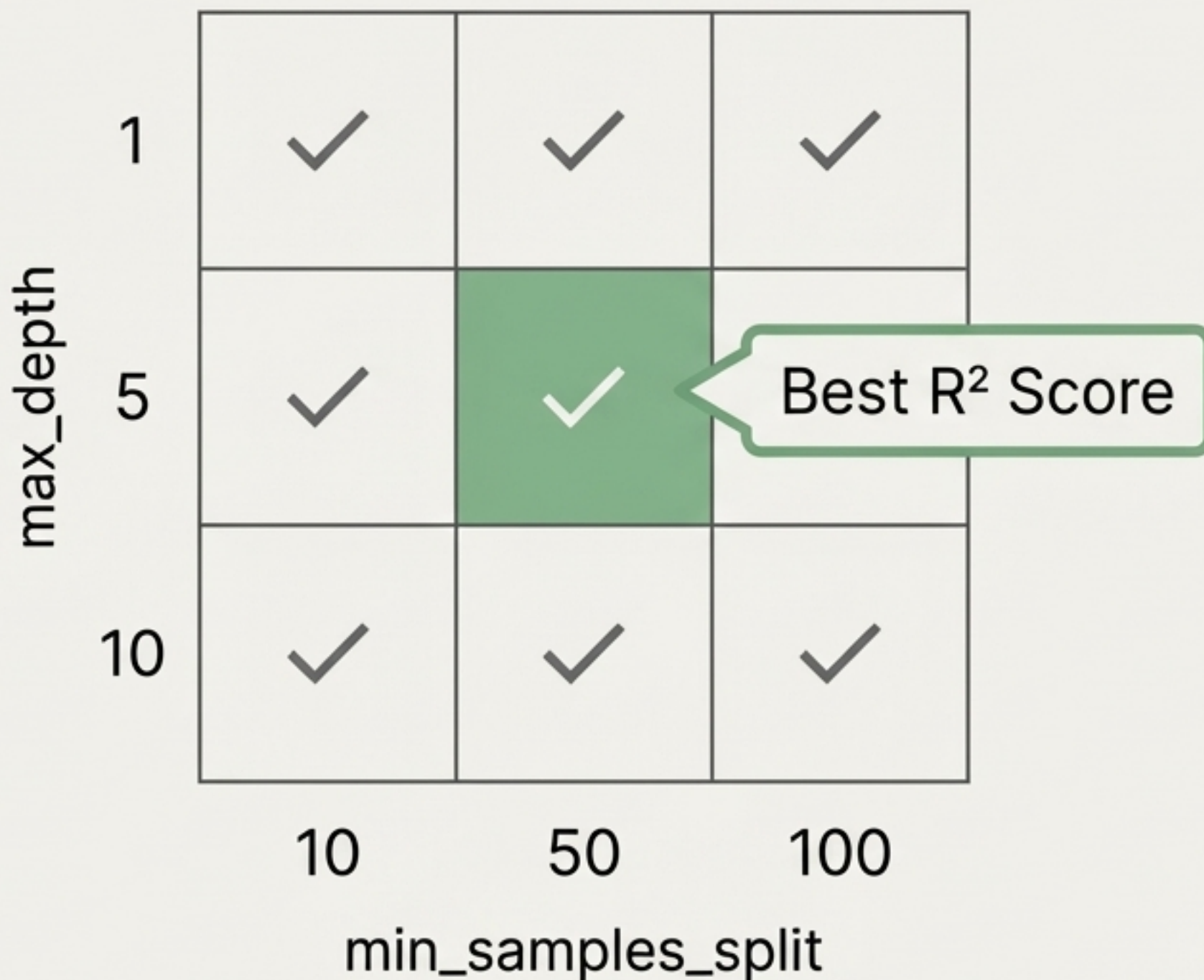


The Dimension Limit (`max_features`)

Restricts the number of columns considered at each split. Crucial for high-dimensionality datasets to reduce variance.

“Automating the Search” in Playfair Display

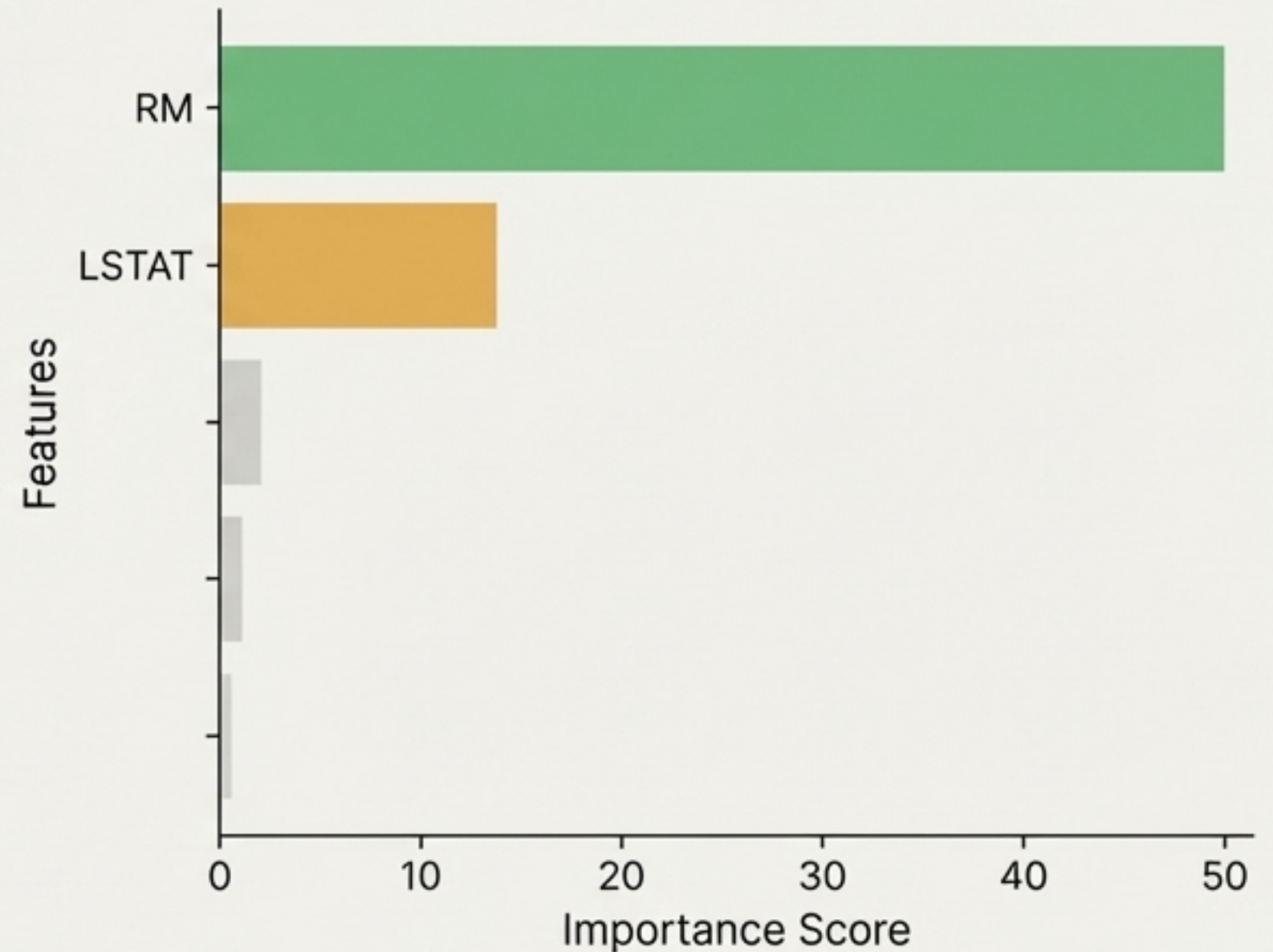
You don't guess hyperparameters; you calculate them. Using tools like GridSearchCV or RandomizedSearchCV, we systematically iterate through combinations of parameters on the dataset (like the Boston Housing data) to mathematically pinpoint the optimal tuning.



Uncovering Feature Importance

Regression Trees inherently rank your data. The `feature_importances_` attribute measures how effectively each specific column reduces the Mean Squared Error across all its splits.

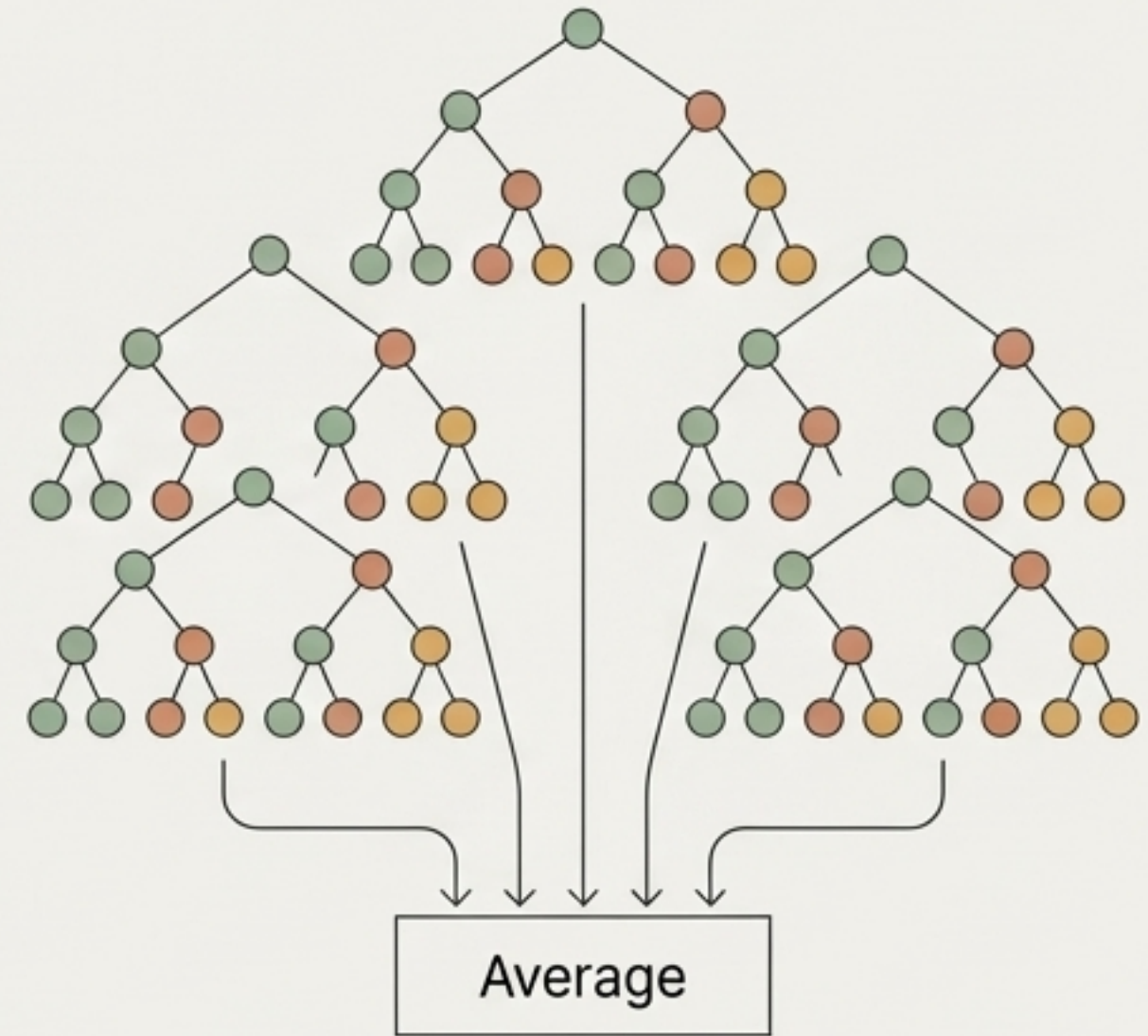
Pro Tip: Use this to conduct dynamic feature selection. Drop the columns that contribute nothing to the model's accuracy.



Evolution: Trees vs. Forests



Single Decision Tree



Random Forest Ensemble

A single decision tree is powerful but highly sensitive to noise and prone to overfitting. For production environments, we combine hundreds of trees into a Random Forest ensemble. It utilises the exact same mechanics, but averages the predictions for unparalleled stability.

The Regression Tree Cheat Sheet

Core Mechanics

- Predicts using the **Mean** of the leaf.
- Splits by minimising **Mean Squared Error** (MSE).
- Builds **step-functions**, not lines.

Key Hyperparameters

- **max_depth** (Height cap)
- **min_samples_split** (Node limit)
- **min_samples_leaf** (End-point limit)
- **splitter** (Best vs. Random)

Practical Usage

- Perfect for **non-linear** data.
- Easily **interpreted**.
- Automatically yields **feature_importances_**.
- Prone to **overfitting** without strict hyperparameter control.